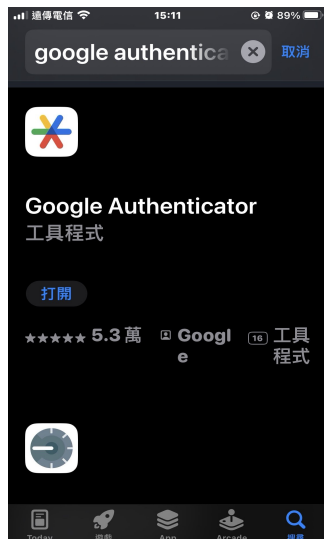


Spring Boot 3.x Google Authenticator 二步驗證學習步驟

手機下載 Google Authenticator 軟體



1. 介紹 Google Authenticator

Google Authenticator 是一種基於時間的一次性密碼（TOTP）驗證機制，可用於增強應用程式的安全性。使用者需透過 Google Authenticator App 掃描 QR Code，並輸入動態產生的驗證碼以完成登入。

2. 建立 Spring Boot 專案

2.1 新增專案

使用 Spring Initializr 建立專案，並選擇以下依賴：

- Spring Web
- Thymeleaf（若需前端頁面）

2.2 引入必要的相依性

在 pom.xml 中加入 Google Authenticator 和 ZXing 相關的 Maven 依賴：

```
<dependencies>
  <dependency>
    <groupId>com.warrenstrange</groupId>
```

```

        <artifactId>googleauth</artifactId>
        <version>1.4.0</version>
    </dependency>
    <dependency>
        <groupId>com.google.zxing</groupId>
        <artifactId>core</artifactId>
        <version>3.4.1</version>
    </dependency>
    <dependency>
        <groupId>com.google.zxing</groupId>
        <artifactId>javase</artifactId>
        <version>3.4.1</version>
    </dependency>
</dependencies>

```

3. 建立 Google Authenticator 服務

在 `GoogleAuthController.java` 中實作：

- 產生密鑰 (Secret Key)
- 產生 QR Code
- 驗證 OTP

3.1 產生密鑰

```

@GetMapping("/generate-key")
public String generateKey() {
    GoogleAuthenticatorKey key = gAuth.createCredentials();
    return "你的密鑰 (Secret Key): " + key.getKey();
}

```

3.2 產生 QR Code

```

@GetMapping("/generate-qr")
public ModelAndView generateQRCode(Model model) throws WriterException
{
    GoogleAuthenticatorKey key = gAuth.createCredentials();
    String otpAuthURL = "otpauth://totp/MyApp?secret=" + key.getKey() +
"&issuer=MyApp";
}

```

```

        String qrCodeBase64 = generateQRCodeImage(otpAuthURL);
        model.addAttribute("qrCode", qrCodeBase64);
        model.addAttribute("secret", key.getKey());
        return new ModelAndView("qrview");
    }
}

```

qrview.html

```

<!DOCTYPE html>
<html lang="zh-tw" xmlns:th="http://www.thymeleaf.org">
<head>
    <meta charset="UTF-8">
    <title>Google Authenticator QR Code</title>
</head>
<body>
    <h2>掃描 QR Code 以設定 Google Authenticator</h2>
    
    <h2>
    <p>密鑰 (Secret Key) : <label th:text="${ secret }"></label></p>
        User Name : <input type="text" value="teacher"
style="font-size:20px"/><p>
        Phone : <input type="text" value="23826015"
style="font-size:20px"/>
    <p></p>
        (Secret Key),User Name,Phone 必須存入資料庫
    </h2>
</body>
</html>

```

3.3 驗證 OTP

```

@PostMapping("/validate")
public String validateCode(@RequestParam String secret, @RequestParam
int code) {
    boolean isCodeValid = gAuth.authorize(secret, code);
    return isCodeValid ? "驗證成功 !" : "驗證失敗 !";
}

```

4. 測試驗證功能

1. 啟動 Spring Boot 伺服器。
2. 訪問 `/auth/generate-qr` 來產生 QR Code。
3. 使用 Google Authenticator 掃描 QR Code。
4. 在 `/auth/validate` API 提交 OTP 來驗證登入。

測試驗證需要之登入頁面及登入驗證 **SecretKey="YEAFLPGDKHFEFZN";**

```
@GetMapping("/login")
public ModelAndView loginPage() {
    return new ModelAndView("login");
}

@PostMapping("/login")
public String login(@RequestParam String secret, @RequestParam int code,
Model model) {

    boolean isCodeValid = gAuth.authorize(secret, code);
    if (isCodeValid) {
        model.addAttribute("message", "登入成功！");
        return "登入成功";
    } else {
        model.addAttribute("message", "驗證失敗，請重試。");
        return "驗證失敗";
    }
}

<!DOCTYPE html>
<html lang="zh-tw" xmlns:th="http://www.thymeleaf.org">
<head>
    <meta charset="UTF-8">
    <title>登入</title>
</head>
<body>
    <h2>登入</h2>
    <form action="/auth/login" method="post">
        <input type="hidden" name="secret" value="YEAFLPGDKHFEFZN">
        <label>Google 手機驗證碼：</label>
        <input type="text" name="code" required><br>
        <button type="submit">登入</button>
    </form>

</body>
</html>
```

完整程式碼:

```
import com.google.zxing.BarcodeFormat;
import com.google.zxing.WriterException;
import com.google.zxing.common.BitMatrix;
import com.google.zxing.qrcode.QRCodeWriter;
import com.warrenstrange.googleauth.GoogleAuthenticator;
import com.warrenstrange.googleauth.GoogleAuthenticatorKey;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.servlet.ModelAndView;
import org.springframework.ui.Model;

import java.awt.image.BufferedImage;
import javax.imageio.ImageIO;
import java.io.ByteArrayOutputStream;
import java.util.Base64;

@RestController
@RequestMapping("/auth")
public class GoogleAuthController {
    String SecretKey="YEAKFLPGDKHFEFZN";

    private final GoogleAuthenticator gAuth = new GoogleAuthenticator();

    @GetMapping("/login")
    public ModelAndView loginPage() {
        return new ModelAndView("login");
    }

    @PostMapping("/login")
    public String login(@RequestParam String secret, @RequestParam int code,
Model model) {
        //固定測試 key
        secret= SecretKey;
        boolean isCodeValid = gAuth.authorize(secret, code);
        if (isCodeValid) {
            model.addAttribute("message", "登入成功！");
            return "登入成功";
        }
    }
}
```

```

        } else {
            model.addAttribute("message", "驗證失敗，請重試。");
            return "驗證失敗";
        }
    }

    @GetMapping("/generate-key")
    public String generateKey() {
        GoogleAuthenticatorKey key = gAuth.createCredentials();
        return "你的密鑰 (Secret Key): " + key.getKey();
    }

//    @GetMapping("/generate-qr")
//    public String generateQRCode() throws WriterException {
//        GoogleAuthenticatorKey key = gAuth.createCredentials();
//        String otpAuthURL = "otpauth://totp/JavaWebApp?secret=" +
key.getKey() + "&issuer=MyApp";
//        return generateQRCodeImage(otpAuthURL);
//    }

    @PostMapping("/validate")
    public String validateCode(@RequestParam String secret, @RequestParam int
code) {
        boolean isCodeValid = gAuth.authorize(secret, code);
        return isCodeValid ? "驗證成功！" : "驗證失敗！";
    }

    @GetMapping("/generate-qr")
    public ModelAndView generateQRCode(Model model) throws
WriterException {
        GoogleAuthenticatorKey key = gAuth.createCredentials();
        String otpAuthURL = "otpauth://totp/Vincent App?secret=" + key.getKey()
+ "&issuer=Login Check";
        String qrCodeBase64 = generateQRCodeImage(otpAuthURL);
        model.addAttribute("qrCode", qrCodeBase64);
        model.addAttribute("secret", key.getKey());

        return new ModelAndView("qrview");
    }

```

```

    }

    private String generateQRCodeImage(String data) throws WriterException {
        QRCodeWriter qrCodeWriter = new QRCodeWriter();
        BitMatrix bitMatrix = qrCodeWriter.encode(data, BarcodeFormat.QR_CODE,
250, 250);
        BufferedImage image = new BufferedImage(250, 250,
BufferedImage.TYPE_INT_RGB);
        for (int x = 0; x < 250; x++) {
            for (int y = 0; y < 250; y++) {
                image.setRGB(x, y, bitMatrix.get(x, y) ? 0x000000 : 0xFFFFFF);
            }
        }
        try {
            ByteArrayOutputStream baos = new ByteArrayOutputStream();
            ImageIO.write(image, "png", baos);
            return "data:image/png;base64," +
Base64.getEncoder().encodeToString(baos.toByteArray());
        } catch (Exception e) {
            return "無法產生 QR Code";
        }
    }
}

```

qrview.html

```

<!DOCTYPE html>
<html lang="zh-tw" xmlns:th="http://www.thymeleaf.org">
<head>
    <meta charset="UTF-8">
    <title>Google Authenticator QR Code</title>
</head>
<body>
    <h2>掃描 QR Code 以設定 Google Authenticator</h2>
    
    <h2>
    <p>密鑰 (Secret Key) : <label th:text="${ secret }"></label></p>
    User Name : <input type="text" value="teacher" style="font-size:20px"/><p>

```

```
Phone : <input type="text" value="23826015" style="font-size:20px"/>
<p></p>
(Secret Key),User Name,Phone 必須存入資料庫
</h2>
</body>
</html>
login.html
<!DOCTYPE html>
<html lang="zh-tw" xmlns:th="http://www.thymeleaf.org">
<head>
  <meta charset="UTF-8">
  <title>登入</title>
</head>
<body>
  <h2>登入</h2>
  <form action="/auth/login" method="post">
    <input type="hidden" name="secret" value="YEAKFLPGDKHFEFZN">
    <label>Google 手機驗證碼：</label>
    <input type="text" name="code" required><br>
    <button type="submit">登入</button>
  </form>

</body>
</html>
```